

Table 1. compares the prime-counting error at powers of ten, the difference between $\text{Li}(10^n)$ and $\pi(10^n)$, and the candidate term $\text{Ccand}(n)$ proposed in this study, for $n = 1\text{--}25$.

10^n	$\pi(10^n)$	$\text{C_cand}(n)$	$\pi(10^n)-4$	C_comp	$\text{Li}-\pi$
10^1	4	0	0	0	2
10^2	25	$(8\cancel{3})(10^1-1)$	21	3	5
10^3	168	$(8\cancel{3})(10^2-1)$	164	100	10
10^4	1,229	$(8\cancel{3})(10^3-1)$	1,225	1,439	17
10^5	9,592	$(8\cancel{3})(10^4-1)$	9,588	17,076	38
10^6	78,498	$(8\cancel{3})(10^5-1)$	78,494	188,170	130
10^7	664,579	$(8\cancel{3})(10^6-1)$	664,575	2,002,089	339
10^8	5,761,455	$(8\cancel{3})(10^7-1)$	5,761,451	20,905,213	754
10^9	50,847,534	$(8\cancel{3})(10^8-1)$	50,847,530	215,819,134	1,701
10^{10}	455,052,511	$(8\cancel{3})(10^9-1)$	455,052,507	2,211,614,157	3,103
10^{11}	4,118,054,813	$(8\cancel{3})(10^{10}-1)$	4,118,054,809	22,548,611,855	11,588
10^{12}	37,607,912,018	$(8\cancel{3})(10^{11}-1)$	37,607,912,014	229,058,754,650	38,263
10^{13}	346,065,536,839	$(8\cancel{3})(10^{12}-1)$	346,065,536,835	2,320,601,129,829	108,971
10^{14}	3,204,941,750,802	$(8\cancel{3})(10^{13}-1)$	3,204,941,750,798	23,461,724,915,866	314,890
10^{15}	29,844,570,422,669	$(8\cancel{3})(10^{14}-1)$	29,844,570,422,665	236,822,096,243,999	1,052,619
10^{16}	279,238,341,033,925	$(8\cancel{3})(10^{15}-1)$	279,238,341,033,921	2,387,428,325,632,743	3,214,632
10^{17}	2,623,557,157,654,233	$(8\cancel{3})(10^{16}-1)$	2,623,557,157,654,229	24,043,109,509,012,435	7,956,589
10^{18}	24,739,954,287,740,860	$(8\cancel{3})(10^{17}-1)$	24,739,954,287,740,856	241,926,712,378,925,808	7,927,346
10^{19}	234,057,667,276,344,607	$(8\cancel{3})(10^{18}-1)$	234,057,667,276,344,603	2,432,608,999,390,322,061	99,877,775
10^{20}	2,220,819,602,560,918,840	$(8\cancel{3})(10^{19}-1)$	2,220,819,602,560,918,836	24,445,847,064,105,747,828	4,583,891
10^{21}	21,127,269,486,018,731,928	$(8\cancel{3})(10^{20}-1)$	21,127,269,486,018,731,924	245,539,397,180,647,934,740	1,157,446,844
10^{22}	201,467,286,689,315,906,290	$(8\cancel{3})(10^{21}-1)$	201,467,286,689,315,906,286	2,465,199,379,977,350,760,378	193,519,524
10^{23}	1,925,320,391,606,803,968,923	$(8\cancel{3})(10^{22}-1)$	1,925,320,391,606,803,968,919	24,741,346,275,059,862,697,745	1,215,343,115
10^{24}	18,435,599,767,349,200,867,866	$(8\cancel{3})(10^{23}-1)$	18,435,599,767,349,200,867,862	248,231,066,899,317,465,798,802	1,714,694,528
10^{25}	176,846,309,399,143,769,411,680	$(8\cancel{3})(10^{24}-1)$	176,846,309,399,143,769,411,676	2,489,820,357,267,522,897,254,988	5,516,098,882